

AWS EC2 Instance

Login & SSH Access Guide

PuTTY | MobaXterm | Termux | CMD | PowerShell | Linux | macOS | EC2 Console

Prerequisites – Before You Begin

Before logging into any EC2 instance, ensure you have the following:

1. EC2 Instance running (State: Running) in AWS Console
2. Key Pair (.pem file) downloaded when instance was launched
3. Security Group with port 22 (SSH) open — Inbound rule: TCP 22, Source: Your IP
4. Public IP / Public DNS of the instance (from EC2 Console > Instances)
5. Default username based on AMI:
 - Amazon Linux / Amazon Linux 2 → ec2-user
 - Ubuntu → ubuntu
 - CentOS → centos
 - RHEL → ec2-user or root
 - Debian → admin
 - SUSE Linux → ec2-user

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Method 1 — PuTTY (Windows)

PuTTY is a free, open-source SSH client for Windows. It requires converting the .pem key to .ppk format using PuTTYgen.

Step A — Convert .pem to .ppk using PuTTYgen

1	Download PuTTY & PuTTYgen from https://www.putty.org
2	Open PuTTYgen (search in Start Menu).
3	Click 'Load' and select your .pem file (change filter to 'All Files (*.*)').
4	PuTTYgen imports the key — click 'Save private key' and save as .ppk file.
5	Click 'Yes' when asked about passphrase (unless you want one).

Step B — Login using PuTTY

1	Open PuTTY application.
2	In 'Host Name (or IP address)' field, enter your EC2 Public IP e.g.: 13.233.xx.xx
3	Port: 22 Connection type: SSH
4	In the left panel, go to: Connection > SSH > Auth > Credentials
5	Click 'Browse' next to 'Private key file for authentication' and select your .ppk file.
6	Go back to 'Session' in left panel, enter a name in 'Saved Sessions', click 'Save'.
7	Click 'Open' to start the connection.
8	Accept the host key fingerprint (click 'Accept' / 'Yes' in the security alert).
9	At the login prompt, type the username: ec2-user (or ubuntu, centos, etc.)
10	Press Enter — you are now logged in!

Tip: Save your session in PuTTY so you don't have to re-enter settings every time.

Note: Never share your .pem or .ppk file. Keep it secured.

Method 2 — MobaXterm (Windows)

MobaXterm is an enhanced terminal for Windows with built-in SSH client, X11 server, and many network tools. It directly accepts .pem files without conversion.

Login Steps

1	Download MobaXterm from https://mobaxterm.mobatek.net (Home Edition is free).
2	Open MobaXterm and click 'Session' button in the top-left.
3	Select 'SSH' from the session type icons.
4	In 'Remote host' field, enter your EC2 Public IP address.
5	Check 'Specify username' and enter: <code>ec2-user</code> (or <code>ubuntu</code> , etc.)
6	Port: 22 (default).
7	Click the 'Advanced SSH settings' tab.
8	Check 'Use private key' and browse to your .pem file.
9	Click 'OK' — MobaXterm connects and logs you in automatically.
10	MobaXterm also opens an SFTP panel on the left to drag/drop files easily.

Tip: MobaXterm saves sessions automatically. Use 'User sessions' tab to find saved connections.

Pro: MobaXterm supports multiple tabs, macros, port tunneling, and built-in file browser — all in one tool.



Method 3 — Termux (Android Mobile)

Termux is a terminal emulator for Android that provides a Linux environment. You can SSH into EC2 directly from your Android phone.

Setup & Login Steps

1	Install Termux from F-Droid (https://f-droid.org) — prefer F-Droid over Play Store for latest version.
2	Open Termux and update packages: <code>pkg update && pkg upgrade</code>
3	Install OpenSSH: <code>pkg install openssh</code>
4	Transfer your .pem key to your Android device (email, Google Drive, or USB).
5	Copy the .pem file to Termux storage. Run: <code>termux-setup-storage</code>
6	Copy key: <code>cp /sdcard/Download/mykey.pem ~/mykey.pem</code>
7	Set correct permissions on the key file: <code>chmod 400 ~/mykey.pem</code>
8	Run SSH command: <code>ssh -i ~/mykey.pem ec2-user@<YOUR-EC2-PUBLIC-IP></code>
9	Type 'yes' to accept fingerprint, and you're connected!

Example: `ssh -i ~/mykey.pem ubuntu@13.233.45.67`

Note: `chmod 400` is mandatory — SSH will refuse connection if key permissions are too open.



Method 4 — Windows Command Prompt (CMD)

Windows 10 (version 1809+) and Windows 11 include a built-in OpenSSH client. No additional software needed.

Check if OpenSSH is Available

```
ssh -V
```

If you see a version number, OpenSSH is ready. If not, install via: Settings > Apps > Optional Features > Add 'OpenSSH Client'.

Login Steps

1	Open Command Prompt (press Win + R, type cmd, press Enter).
2	Navigate to where your .pem file is stored, e.g.: cd C:\Users\YourName\Downloads
3	Set correct permissions on the .pem file (right-click > Properties > Security > Advanced — remove all except your user).
4	Run the SSH command: ssh -i mykey.pem ec2-user@<YOUR-EC2-PUBLIC-IP>
5	Type 'yes' when prompted for fingerprint confirmation.
6	You are now connected to your EC2 instance!

SSH Command Syntax

```
ssh -i C:\path\to\keyfile.pem username@public-ip-or-dns
```

Example: ssh -i C:\keys\mykey.pem ubuntu@54.123.45.67

Issue: If you get 'WARNING: UNPROTECTED PRIVATE KEY FILE', fix .pem permissions as described in Step 3.

Method 5 — Windows PowerShell

PowerShell on Windows 10/11 also supports OpenSSH natively and is often more flexible than CMD for scripting connections.

Fix Key Permissions (PowerShell Method)

Run these commands in PowerShell to fix .pem file permissions:

```
$path = 'C:\keys\mykey.pem'
icaccls $path /reset
icaccls $path /grant:r "$env:USERNAME:(R) "
icaccls $path /inheritance:r
```

Login Steps

1	Open PowerShell (Win + X > Windows PowerShell or Terminal).
2	Run permission fix commands above if first time.
3	Run SSH command: <code>ssh -i C:\keys\mykey.pem ec2-user@<YOUR-EC2-PUBLIC-IP></code>
4	Accept the fingerprint prompt by typing 'yes'.
5	You are connected!

Tip: Add `-o StrictHostKeyChecking=no` to skip fingerprint prompt (not recommended for production).

Method 6 — Linux Terminal (Ubuntu / CentOS / Fedora)

Linux comes with SSH pre-installed. This is the most straightforward method — just set permissions and connect.

Login Steps

1	Open Terminal (Ctrl + Alt + T on Ubuntu).
2	Navigate to your .pem file location: <code>cd ~/Downloads</code>
3	Set correct permissions: <code>chmod 400 mykey.pem</code>
4	Run SSH command: <code>ssh -i mykey.pem ec2-user@<YOUR-EC2-PUBLIC-IP></code>
5	Type 'yes' to accept the fingerprint.
6	You are now logged into your EC2 instance!

Common SSH Options

-i <keyfile>	Specify identity (private key) file
-p <port>	Connect to specific port (default 22)
-v	Verbose output for debugging
-L <local>:<host>:<port>	Local port forwarding (tunneling)
-N	No remote command (for tunneling only)
-o StrictHostKeyChecking=no	Skip host key check (use carefully)

Quick One-liner

```
ssh -i ~/Downloads/mykey.pem ubuntu@13.233.45.67
```

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Method 7 — macOS Terminal

macOS includes OpenSSH by default. The process is nearly identical to Linux.

Login Steps

1	Open Terminal (Applications > Utilities > Terminal, or Cmd + Space, type 'Terminal').
2	Move .pem file to a safe location: <code>mv ~/Downloads/mykey.pem ~/.ssh/</code>
3	Set permissions: <code>chmod 400 ~/.ssh/mykey.pem</code>
4	Connect: <code>ssh -i ~/.ssh/mykey.pem ec2-user@<YOUR-EC2-PUBLIC-IP></code>
5	Type 'yes' to accept fingerprint.
6	You are connected to your EC2 instance.

Add Key to SSH Agent (Optional)

To avoid specifying `-i` every time:

```
ssh-add ~/.ssh/mykey.pem  
ssh ec2-user@<YOUR-EC2-PUBLIC-IP>
```

Tip: On macOS Ventura+, add 'UseKeychain yes' and 'AddKeysToAgent yes' in `~/.ssh/config` for automatic key loading.

Method 8 — AWS EC2 Instance Connect (Browser)

No SSH key or local software needed — connect directly from AWS Console in your web browser.

Login Steps

1	Log in to AWS Console at https://console.aws.amazon.com
2	Navigate to EC2 > Instances.
3	Select your running instance.
4	Click 'Connect' button at the top.
5	Choose 'EC2 Instance Connect' tab.
6	Enter the username (e.g., ec2-user or ubuntu).
7	Click 'Connect' — a browser-based terminal opens!
8	You are now connected with no keys or software required.

Requirement: EC2 Instance Connect requires the aws-ec2-instance-connect package installed on the instance and port 22 open in the Security Group.

Limitation: Session times out after inactivity. Not suitable for long-running sessions.

Method 9 — AWS Systems Manager Session Manager

Connect without opening port 22 at all — no SSH key, no public IP needed. Requires SSM Agent on the instance and an IAM role.

Login Steps

1	Ensure SSM Agent is installed on EC2 (pre-installed on Amazon Linux 2, Ubuntu 20.04+).
2	Attach IAM role with 'AmazonSSMManagedInstanceCore' policy to your EC2 instance.
3	Go to AWS Console > Systems Manager > Session Manager.
4	Click 'Start Session'.
5	Select your EC2 instance from the list.
6	Click 'Start Session' — terminal opens in the browser.

Advantage: Most secure method — no inbound ports needed. All traffic is over HTTPS via AWS.

Method 10 — SSH Config File (All Platforms)

Create an SSH config file to avoid typing long commands every time. Works on Linux, macOS, and Windows (WSL/PowerShell).

Create the Config File

```
nano ~/.ssh/config    (Linux/macOS)
notepad C:\Users\YourName\.ssh\config    (Windows)
```

Config File Contents

```
Host myec2
  HostName      13.233.45.67
  User          ec2-user
  IdentityFile   ~/.ssh/mykey.pem
  Port          22
```

Connect Using Alias

```
ssh myec2
```

Tip: You can add multiple Host entries to manage connections to different EC2 instances.



Troubleshooting Common Issues

Error / Issue	Solution
Permission denied (publickey)	Wrong username or wrong key. Check AMI default user. Verify key pair matches instance.
Connection timed out	Port 22 blocked. Check Security Group inbound rule — TCP port 22 must allow your IP.
WARNING: UNPROTECTED PRIVATE KEY FILE!	Key file permissions too open. Run: <code>chmod 400 keyfile.pem</code> (Linux/macOS) or fix ACL (Windows).
Host key verification failed	Known hosts mismatch. Run: <code>ssh-keygen -R <ip></code> to remove old entry, then reconnect.
Network error: Connection refused	EC2 instance may be stopped, or SSHd is not running. Start instance from console.
Too many authentication failures	SSH is trying too many keys. Add: <code>-o IdentitiesOnly=yes</code> to your command.
ec2-user: Permission denied	Using wrong username. For Ubuntu use 'ubuntu', for RHEL use 'ec2-user' or 'root'.



Quick Reference — SSH Commands by Platform

Linux / macOS / WSL

```
chmod 400 mykey.pem  
ssh -i mykey.pem ec2-user@<PUBLIC-IP>
```

Windows CMD / PowerShell

```
ssh -i C:\keys\mykey.pem ec2-user@<PUBLIC-IP>
```

Termux (Android)

```
chmod 400 ~/mykey.pem  
ssh -i ~/mykey.pem ec2-user@<PUBLIC-IP>
```

PuTTY (Windows)

Convert .pem to .ppk using PuTTYgen, then use PuTTY GUI to connect.

MobaXterm (Windows)

New Session > SSH > Enter IP, username, and browse to .pem file directly.

AWS EC2 Instance Connect (Browser)

AWS Console > EC2 > Instances > Select Instance > Connect > EC2 Instance Connect

AWS Session Manager (No SSH needed)

AWS Console > Systems Manager > Session Manager > Start Session > Select Instance

